

DES Instrument Cluster

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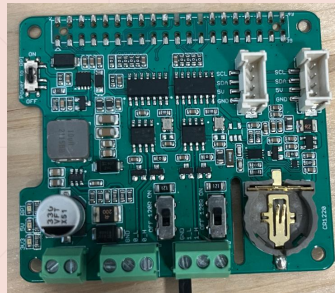
Specification : Hardware

Raspberry Pi

Raspberry Pi 4



**CAN_BUS(FD)
Shield(MCP2518FD)**

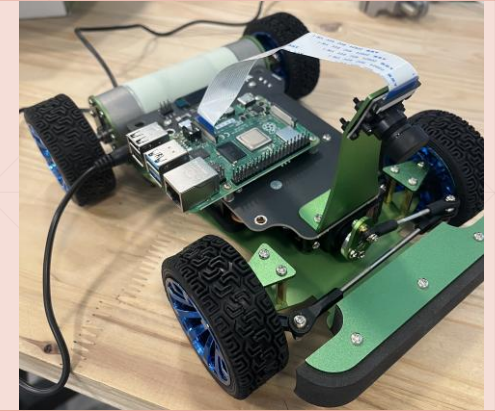


7.9-inch Display



Piracer

Piracer Kit

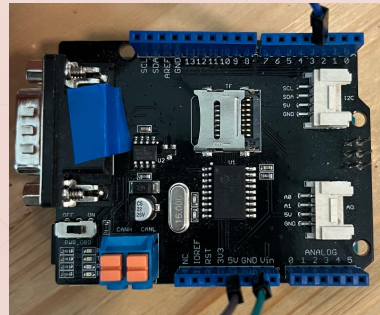


Arduino UNO

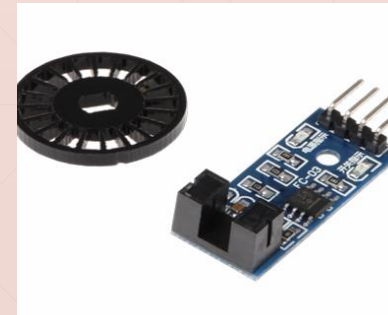
Arduino UNO



**CAN-BUS Shield
(MCP2515)**



**Speed Sensor
(LM 393)**



Joystick

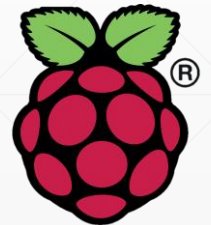


Specification : Software

- Languages : C++, Python, QML



- Frameworks / Libraries : Qt5, dbus-python, piracer gamepads, PyGObject
- IPC (Inter-Process Communication) : D-Bus
- Build System : Qt-based Cross-Compilation (PC -> Raspberry Pi)



Raspberry Pi

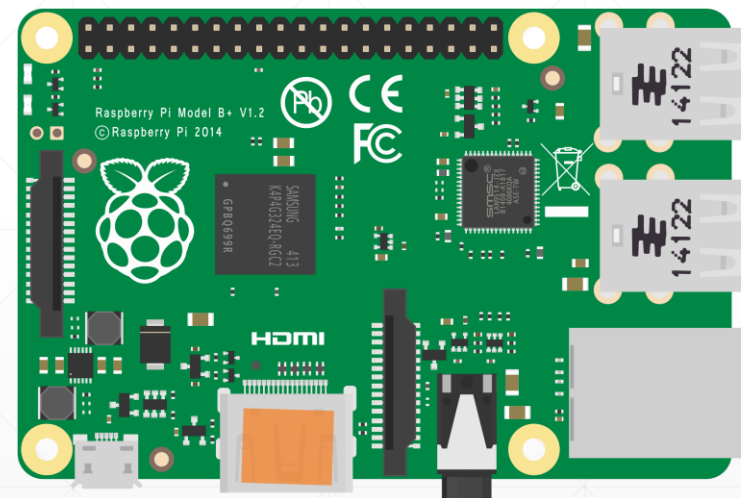
Display



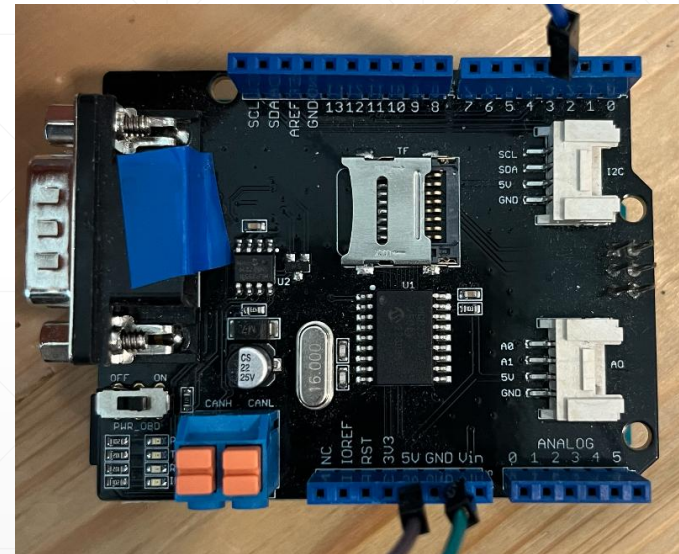
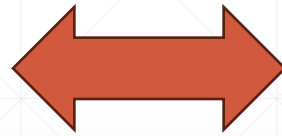
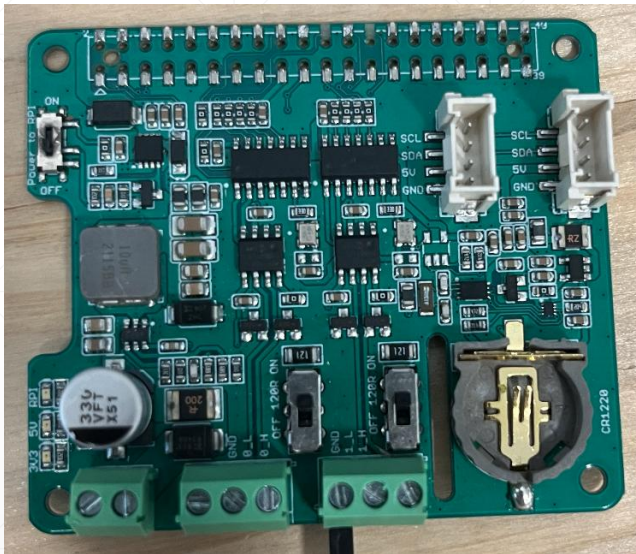
Cross Compile



.exe



CAN Communication



D-Bus

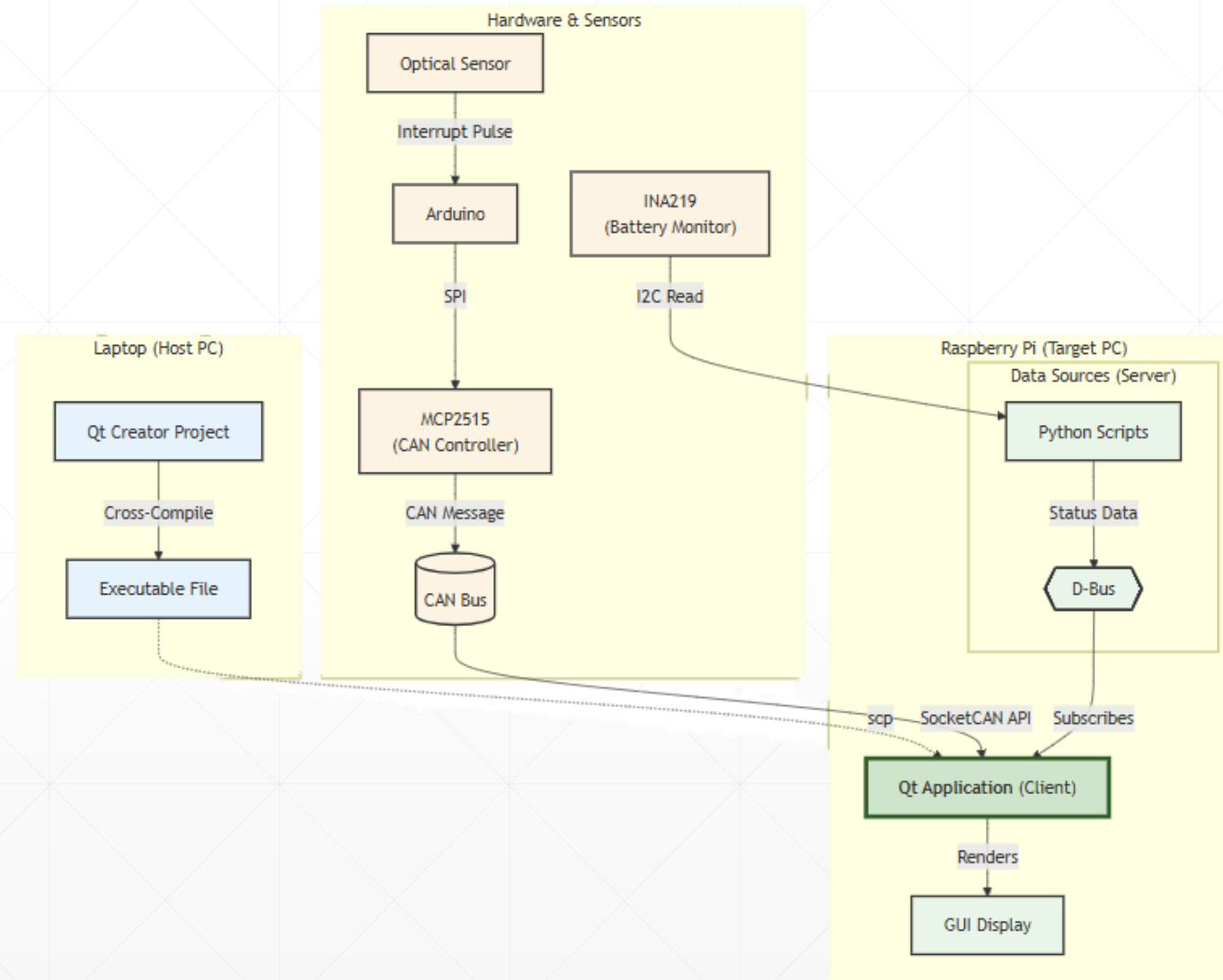


-> Sender (soc.py, vehicle_controller.py)

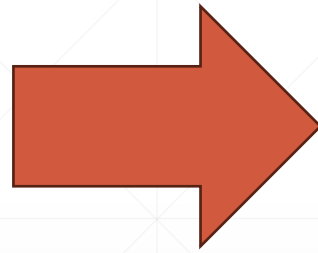


-> Receiver (dbusreceiver.cpp)

Flow



GUI



Thank you

Q&A

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